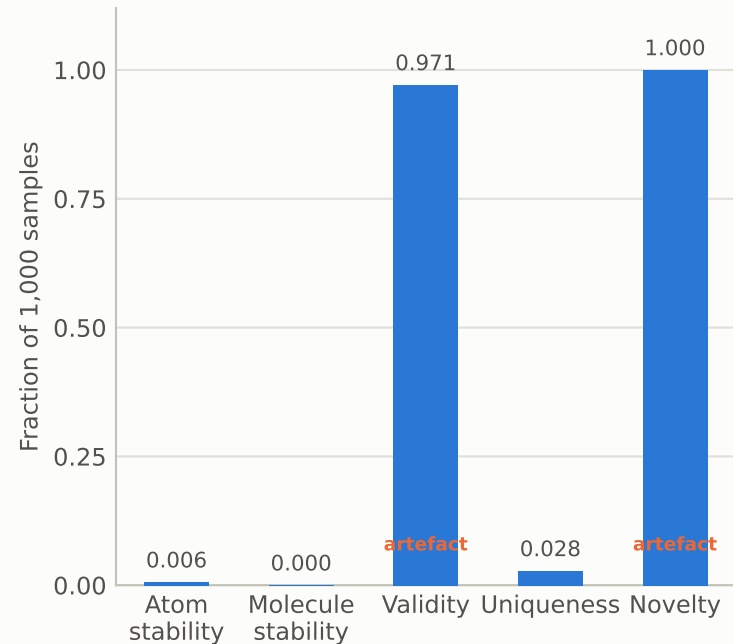
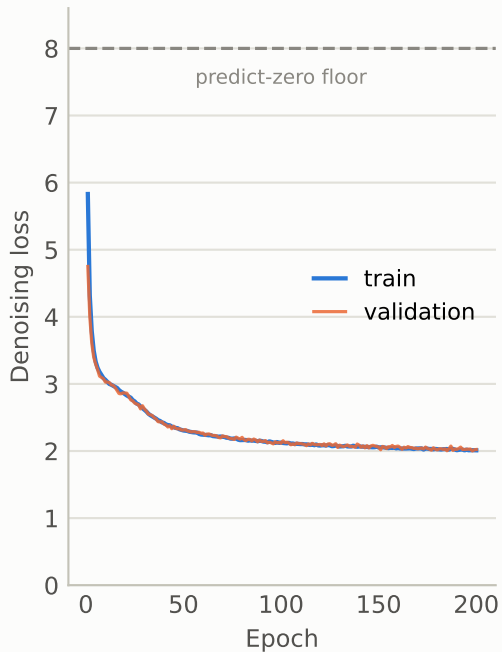


QM9 unconditional generation: MLP denoiser baseline

Sample quality, MLP denoiser on QM9



Training converged



Model setup

denoiser	mlp, 316,904 params
layers x width	4 x 256
data	QM9, 130,831 molecules
split	80 / 10 / 10, seed 0
diffusion	DDPM, T=500, polynomial_2
parameterised	epsilon-prediction, L2
training	200 epochs, batch 64
optimiser	AdamW, lr 0.0002, EMA 0.999
augmentation	random SO(3) per batch
hardware	4 CPU cores, 27 min

Validity and novelty are artefacts: no atom pair lands close enough to bond, so every atom is its own fragment and RDKit sanitises a lone "C".